

Evaluating and Equivalent Expressions

MA.8.NSO.1.3

I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

So far: every law, every kind of base, every kind of exponent.

Today: use all of it at once — and rewrite bases to compare.

Warm-Up

On your own. Four minutes. No notes.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
2. Find the value: $\frac{1}{5^{-2}}$
3. Find the value: $\frac{10^6}{10^4}$
4. Write 64 as a power of 2, of 4, and of 8.

Warm-Up

Answers.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
 $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$

2. Find the value: $\frac{1}{5^{-2}}$ **25**

3. Find the value: $\frac{10^6}{10^4}$ **100**

4. Write 64 as a power of 2, of 4, and of 8.

2^6 , 4^3 , 8^2

Notes

I. Two students, two plans, one value

$$\text{Melanie: } \frac{5^{-2} \cdot 5^5}{5} = \frac{5^3}{5} = 5^2 = 25$$

$$\text{Adonis: } \frac{5^{-2} \cdot 5^5}{5} = \frac{\frac{1}{25} \cdot 3125}{5} = \frac{125}{5} = 25$$

Apply the laws first, evaluate last. The value is the same; the arithmetic is smaller.

Notes

Copy the worked line.

II. Every law, one expression

$$\frac{(6^2)^{-3}}{6^{-5}} = \frac{6^{-6}}{6^{-5}} = 6^{-6 - (-5)} = 6^{-1} = \frac{1}{6}$$

Find the step that broke:

$$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} = \left(\frac{1}{5}\right)^{-8} \quad \text{— wrong: multiplied; should be } \left(\frac{1}{5}\right)^2$$

$$\left(\frac{3}{2}\right)^{-3} = \frac{2^{-3}}{3^{-3}} \quad \text{— wrong: flipped, kept the minus; should be } \left(\frac{2}{3}\right)^3$$

Notes

III. Different bases, same value — rewrite to a common base

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
3	$3^2 = 9$	$3^3 = 27$	$3^4 = 81$
5	$5^2 = 25$	$5^3 = 125$	$5^4 = 625$

$$8^5 = (2^3)^5 = 2^{15}$$

$$9^4 = (3^2)^4 = 3^8$$

$$2^3 \cdot 4^3 = 2^3 \cdot (2^2)^3 = 2^3 \cdot 2^6 = 2^9$$

Notes

IV. Testing equivalence: Xavier's claim

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3} = 2^{-4} \cdot 2^{-3} = 2^{-7}$$

$$\left(\frac{1}{2}\right)^4 \cdot 4^{-1} = 2^{-4} \cdot 2^{-2} = 2^{-6}$$

Same base, then compare exponents.

$2^{-7} \neq 2^{-6}$ — so Xavier is wrong.

Example 1

Evaluate.

$$\left(\frac{3}{4}\right)^{-2} \cdot 3$$

Find the value.

Example 1

Worked.

$$\left(\frac{3}{4}\right)^{-2} \cdot 3 = \left(\frac{4}{3}\right)^2 \cdot 3 \quad \text{negative exponent: flip}$$

$$\frac{16}{9} \cdot 3 = \frac{16}{3} \quad \text{power of a quotient, then multiply}$$

Answer: 16/3

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

Your Turn

Answer.

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

$$(5/2)^2 \cdot 2 = 25/4 \cdot 2$$

$$\textbf{Answer: } \frac{25}{2}$$

Example 2

Equivalent or not?

Are $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$ equivalent?

Rewrite both with the bases 3 and 2, then compare.

Example 2

Worked.

$$3^2 \cdot 6^3 = 3^2 \cdot 3^3 \cdot 2^3 = 3^5 \cdot 2^3 \quad \text{power of a product, product of powers}$$

$$\frac{(3 \cdot 2)^5}{2^2} = \frac{3^5 \cdot 2^5}{2^2} = 3^5 \cdot 2^3 \quad \text{power of a product, quotient of powers}$$

$$3^5 \cdot 2^3 = 3^5 \cdot 2^3 \quad \text{same bases, same exponents}$$

Answer: Yes — both are $3^5 \cdot 2^3 = 1944$

Your Turn

Same steps, your numbers. Boards up when done.

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

Your Turn

Answer.

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

both are $2^5 \cdot 5^3$

Answer: Yes — $2^5 \cdot 5^3 = 4000$

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

Answer it.

Find the value.

Whiteboards • Question 1 of 9

Answer.

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

$(-7)^5$ over $(-7)^2$, so $(-7)^3$

Answer: -343

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

$$-3 - (-2) = -1, \text{ so } (3/4)^{-1}$$

Answer: $\frac{4}{3}$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

$$2^{-3} \cdot 2^4 \text{ over } 2^3 = 2^{-2}$$

$$\textbf{Answer: } \frac{1}{4}$$

Whiteboards • Question 4 of 9

Boards up on three.

$$8^5$$

Answer it.

Write it as a power of 2.

Whiteboards • Question 4 of 9

Answer.

$$8^5$$

$$8 = 2^3, \text{ so } (2^3)^5$$

$$\textbf{Answer: } 2^{15}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$9^4$$

Which expression is equivalent?

A. 3^8

B. 3^6

C. 3^4

D. 3^{16}

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$9^4$$

Which expression is equivalent?

A. 3^8

B. 3^6

C. 3^4

D. 3^{16}

Answer: A — 3^8

Whiteboards • Question 6 of 9

Boards up on three.

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^{-5}

C. 2^7

D. 2^{-12}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^{-5}

C. 2^7

D. 2^{-12}

Answer: A — 2^{-7}

Whiteboards • Question 7 of 9

Boards up on three.

$$2^3 \cdot 4^3$$

Answer it.

Write it as one power of 2.

Whiteboards • Question 7 of 9

Answer.

$$2^3 \cdot 4^3$$

$$4^3 = (2^2)^3 = 2^6$$

Answer: 2^9

Whiteboards • Question 8 of 9

Boards up on three.

$$4^7 \cdot 4^{-2}$$

Answer it.

Write it as one power of 2.

Whiteboards • Question 8 of 9

Answer.

$$4^7 \cdot 4^{-2}$$

$$4^5 = (2^2)^5$$

Answer: 2^{10}

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-7} . Do you agree?

Write your answer in sentences.

Rewrite each with base 2, and say what you find.

Whiteboards • Question 9 of 9

Answer.

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-7} . Do you agree?

2^{-7} and 2^{-6}

Answer: No — the second is 2^{-6} , not 2^{-7}

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Evaluate expressions using properties of exponents — 6T7

Identify equivalent expressions involving exponents II — QDM